



جامعة نيويورك أبوظبي
NYU ABU DHABI

Robotics Lab Report

Course: ENGR-UH 4330| Fall 2025

Experiment: SIR Model – Flu Epidemic Simulation in Simulink

Date: 02/ 09 / 2025

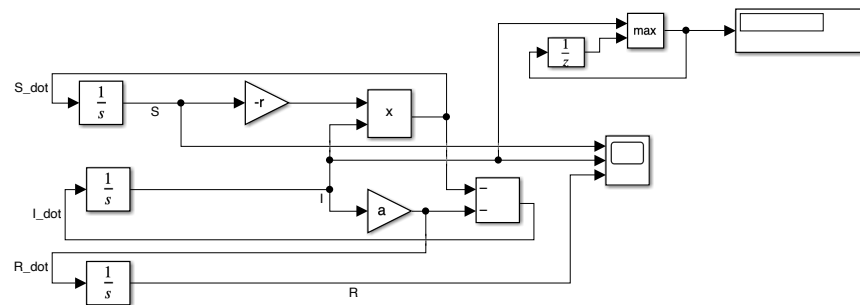
Group Members: Demarce Williams

Objectives

The main objective of this lab is to build and simulate the SIR model in Simulink to understand the dynamics of a flu epidemic. The experiment involves creating a model using the governing differential equations, configuring blocks for integration and display, and analyzing the behavior of the susceptible, infected, and recovered populations over a 14-day period.

Diagram

Figure 1: Simulink Model of SIR Equations



Procedure

1. The system of ODEs for the SIR model was defined:

$$\frac{dS}{dt} = -r * S * I$$

$$\frac{dI}{dt} = r * S * I - a * I$$

$$\frac{dR}{dt} = a * I$$

with $r = 0.00218$, $a = 0.5$, and t in days.

2. A Simulink model was constructed using:

- Integrator blocks to represent accumulation of S, I, and R.
- Gain and Product blocks to model infection and recovery terms.
- A Scope block to visualize S, I, and R curves.
- A Display block to show the maximum number of infected individuals.

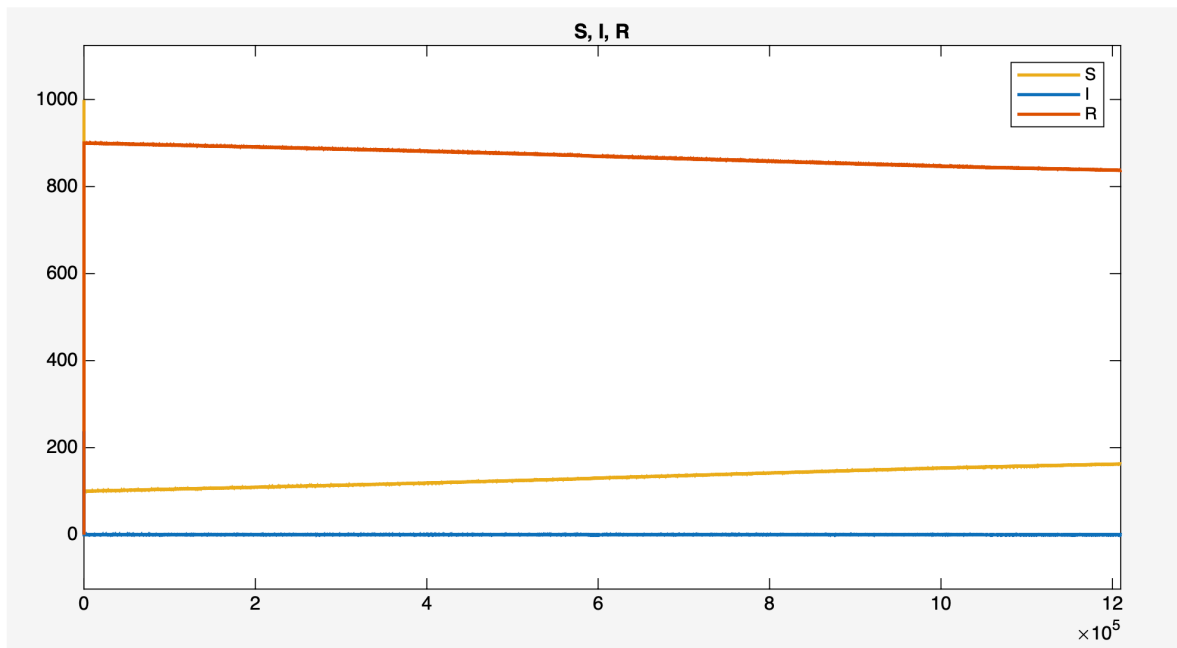
3. The initial conditions were set as follows:

Variable	Initial Value
Total Population (N)	1000
Susceptible (S0)	999
Infected (I0)	1
Recovered (R0)	0

Results & Observations

The simulation was run for 14 days. The following outputs were obtained:

Figure 2: SIR Curves over 14 days (to be inserted).

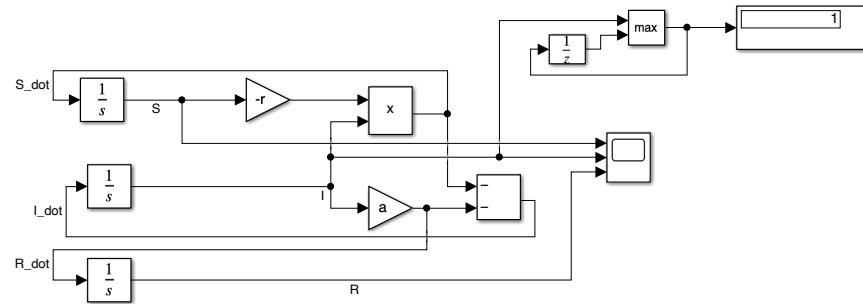


The susceptible population decreases rapidly as the infection spreads.

The infected population rises to a peak before declining.

The recovered population steadily increases as individuals recover from the infection.

Figure 3: Maximum Number of Infected Individuals Displayed



The maximum peak value of the infected was: 1.

Conclusion

This lab successfully demonstrated the implementation of the SIR model in Simulink. The simulation provided insights into how infection and recovery rates affect the dynamics of disease spread. The scope plots showed the interaction between the S, I, and R populations, and the display block highlighted the maximum number of infected individuals during the epidemic.